MANIPAL UNIVERSITY COLLEGE MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

ANATOMY
OSTEOLOGY OF BONY PELVIS



SHARMMA

Anatomy department

FoM, MUCM

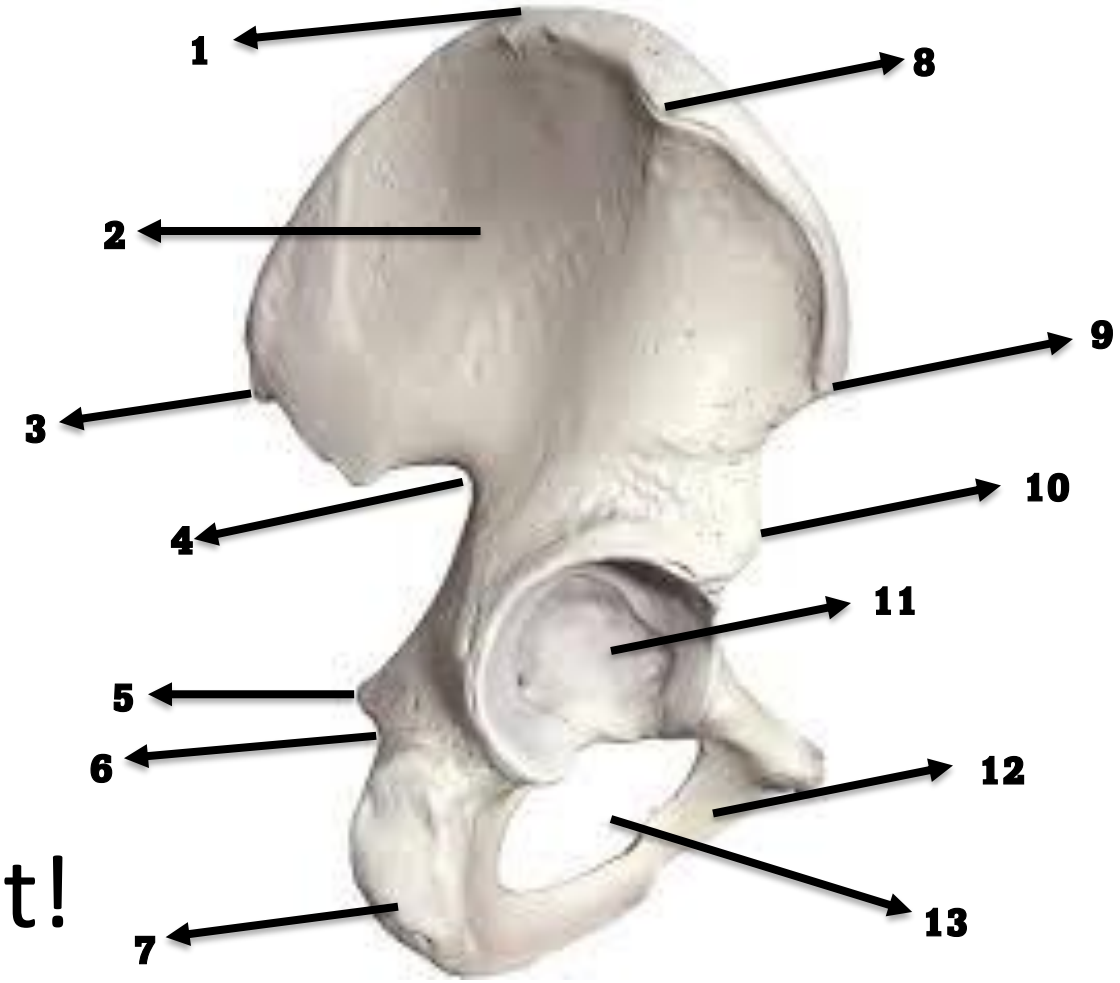
#inspire

Learning Outcomes

By the end of this session, students should be able to:

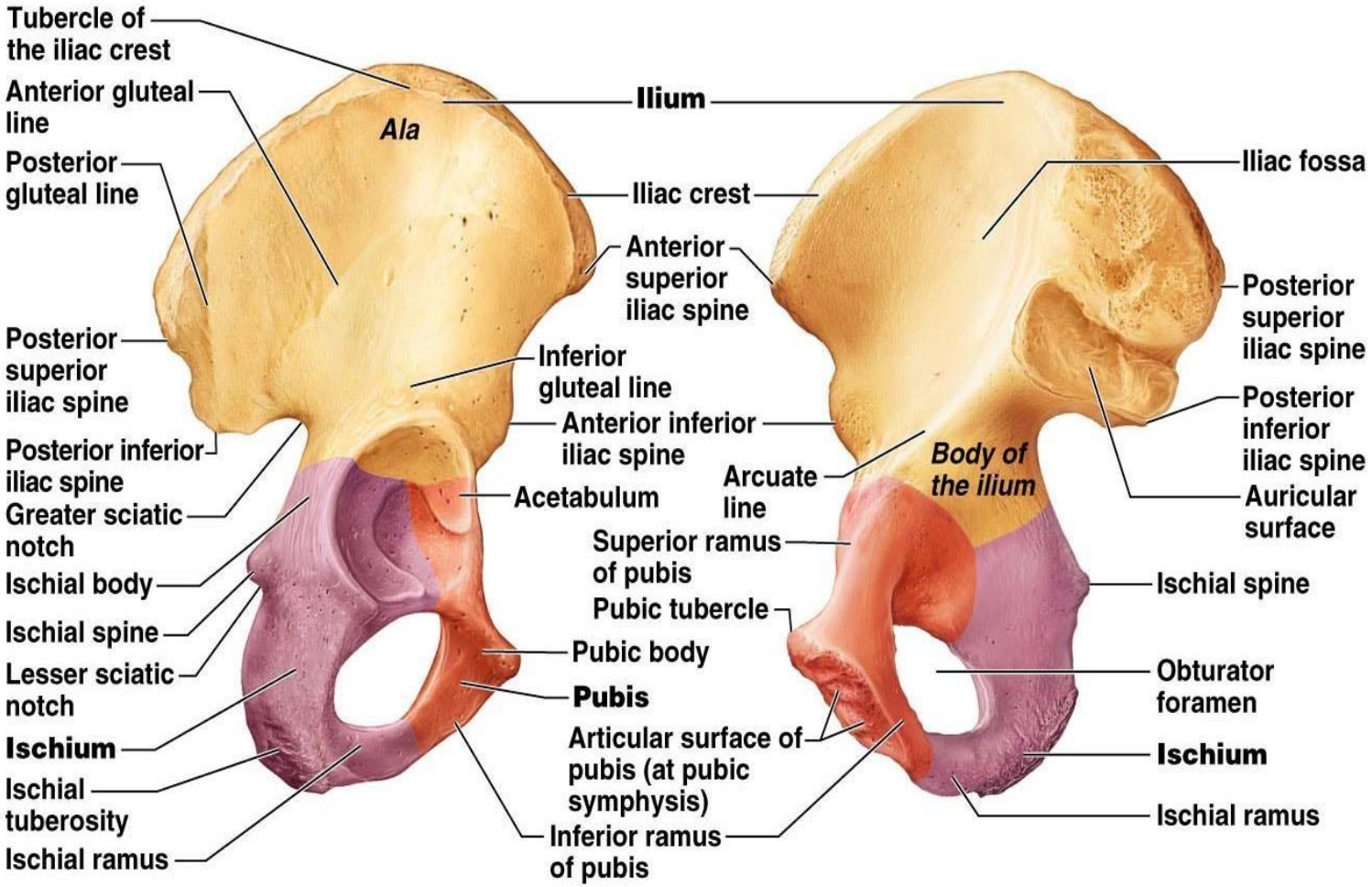
- Identify the bones forming the pelvis
- Describe the joints of the pelvis
- Explain the parts and features of the pelvis
- Differentiate between male and female pelvis
- Apply anatomical knowledge in clinical and forensic contexts

Hip Bone - Recap



Name it!

Summary: Lateral and medial view of hip bone



(b) Lateral view, right hip bone

(c) Medial view, right hip bone

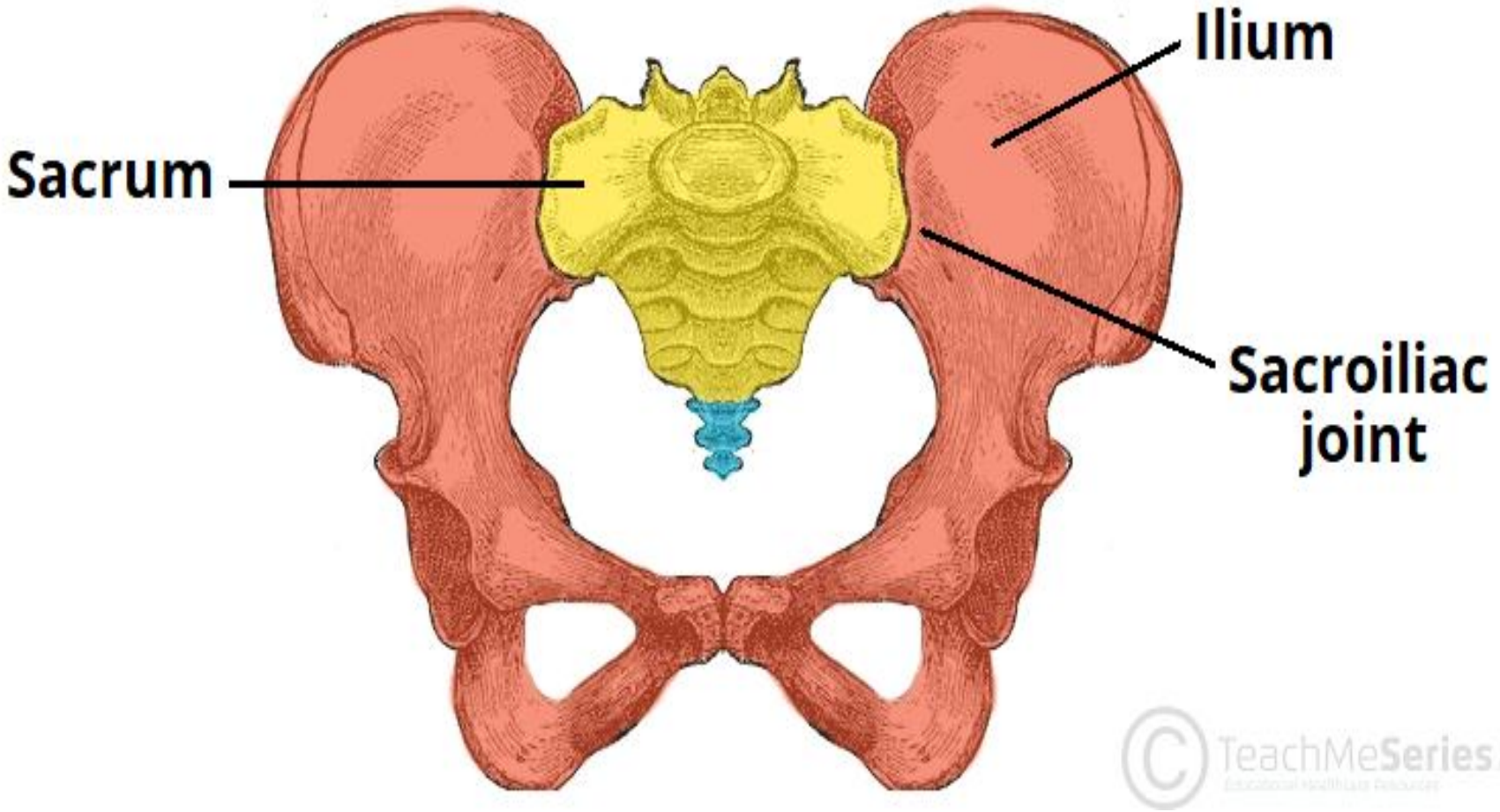
Bones of the Pelvis

Hip bones (Ilium, Ischium, Pubis)

- Sacrum
- Coccyx

- Each hip bone is formed by fusion of ilium, ischium, and pubis.

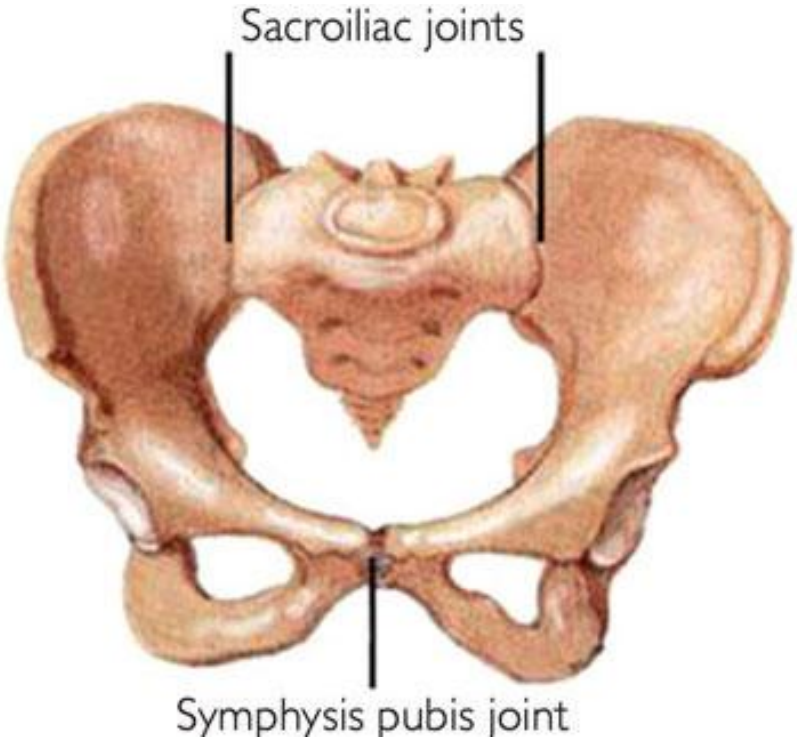
Bones of the Pelvis

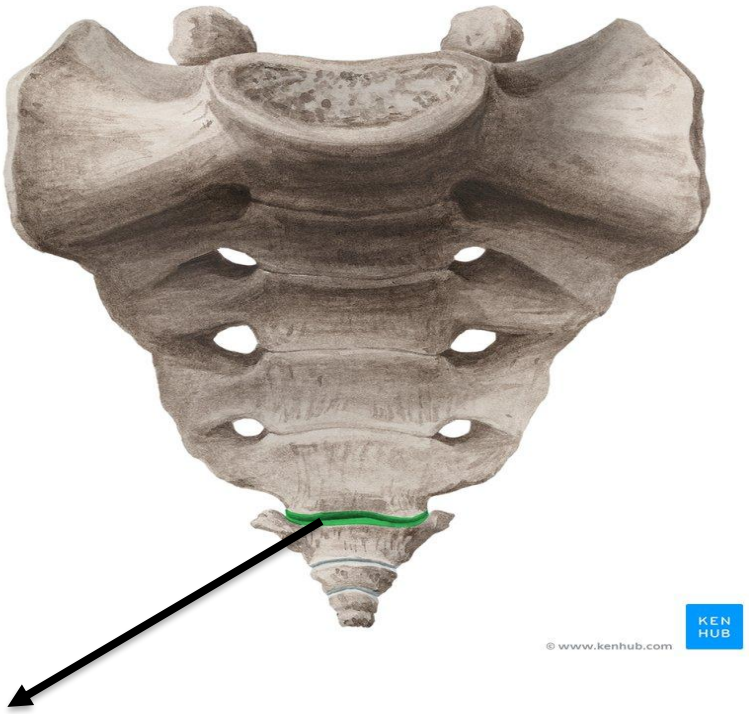


Joints of the Pelvis

- Sacroiliac joints (2)
- Pubic symphysis
- Sacrococcygeal joint

- These provide stability and limited mobility.





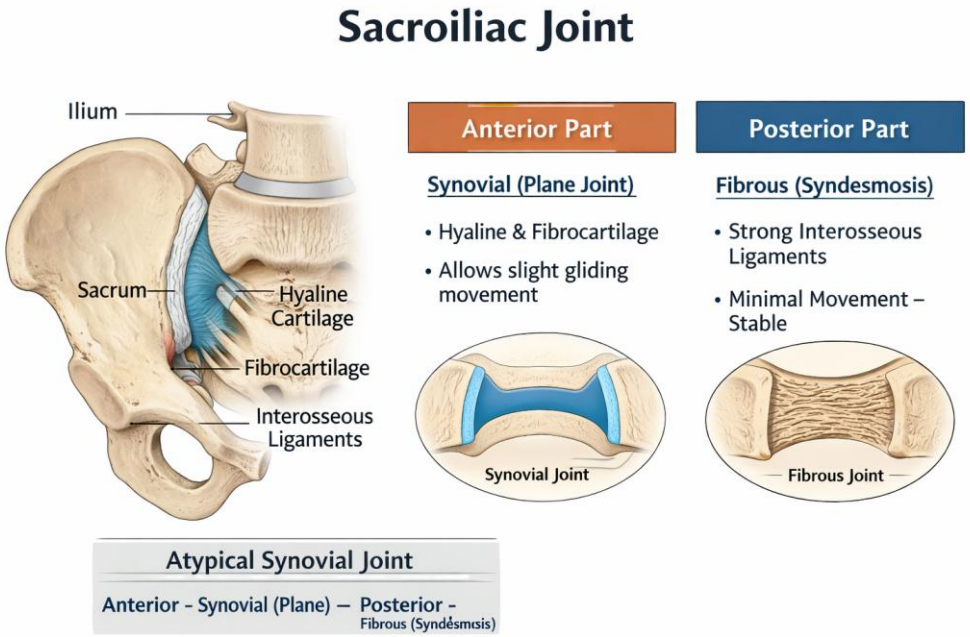
Sacrococcygeal joint

SYMPHYSIS PUBIS

It is a secondary cartilaginous joint between the bodies of two pubic bones. The articular surfaces are covered by a plate of hyaline cartilage which are then united by a disc of fibrocartilage. The joint is surrounded and strengthened by the fibrous ligaments, especially above and below. The **inferior pubic (arcuate) ligament** is very strong.

Sacroiliac joint

- Anterior: plane type of synovial joint between the auricular surfaces of the ilium and sacrum.
- Posterior: syndesmosis between the tuberosities of the sacrum and ilium.



- The **sacrotuberous ligament** is a strong broadband of the fibrous tissue. It has broad upper medial end and narrow lower lateral end. The upper end is attached from above downward to the posterior superior iliac spine, posterior inferior iliac spine, lower part of the posterior surface, and lateral border of the sacrum, and adjoining upper part of the coccyx. Its lower end is attached to the medial margin of the ischial tuberosity. Some of the fibres from the lower end are continued on to the ramus of the ischium to form *falciform process.*
- The **sacrospinous ligament** is a triangular sheet of fibrous tissue, with its apex attached laterally to the ischial spine and base attached medially to the side of sacrum and coccyx.

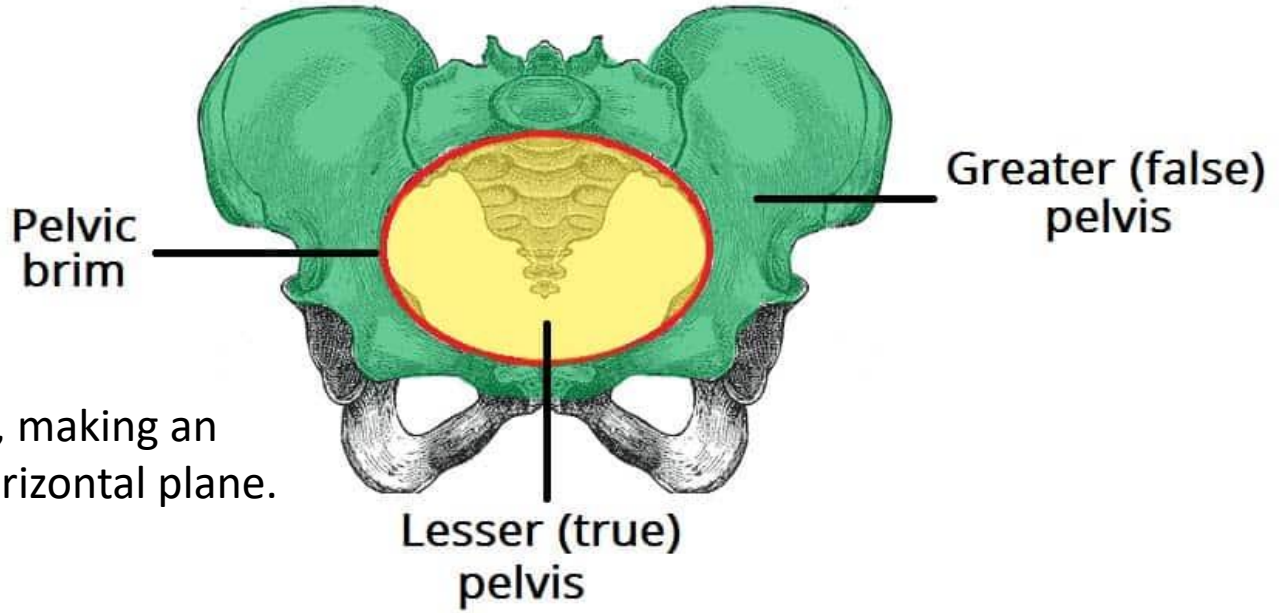
- The **interosseous sacroiliac ligament** is the **second strongest ligament in the body** and connects the irregular bony tuberosities of the sacrum and ilium behind the auricular surfaces.
- The **ventral sacroiliac ligament** connects the ala and pelvic surface of the sacrum with the adjacent part of the ilium.
- The **dorsal sacroiliac ligament** is made up of strong fibrous bands, which connect the posterosuperior iliac spine of ilium with the intermediate sacral crest.
- The **iliolumbar ligament** connects the transverse process of the L5 vertebra to the ilium.

SACROCOCCYGEAL JOINT

It is a secondary cartilaginous joint between the apex of sacrum and the base of coccyx. The bones are united by thin intervertebral disc. The sacrococcygeal joint is strengthened by *ventral and dorsal sacrococcygeal ligaments*, and *intercornual ligaments* (connecting cornua of sacrum and coccyx). The coccyx moves a little backward during defecation and parturition.

Parts of the Pelvis

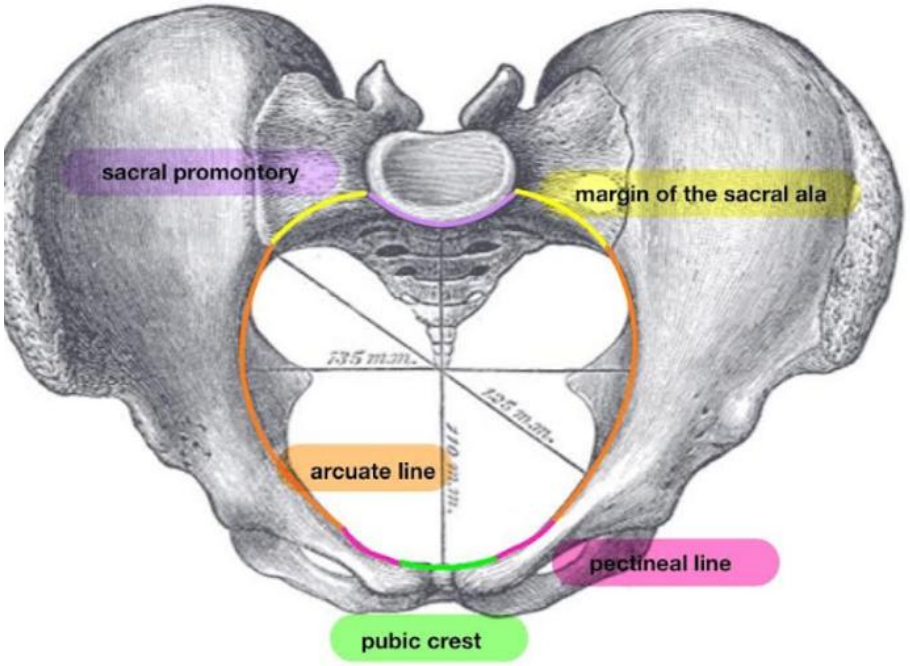
- Greater (false) pelvis – above pelvic brim
- Lesser (true) pelvis – below pelvic brim
- Pelvic inlet and outlet define the boundaries of the true pelvis.

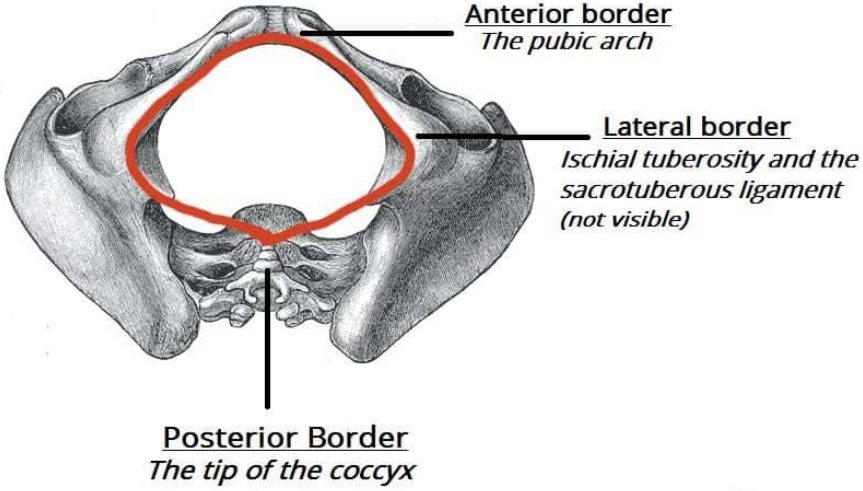


pelvic inlet is an oblique plane, making an angle of 50° to 60° with the horizontal plane.

Bony features of the Pelvis

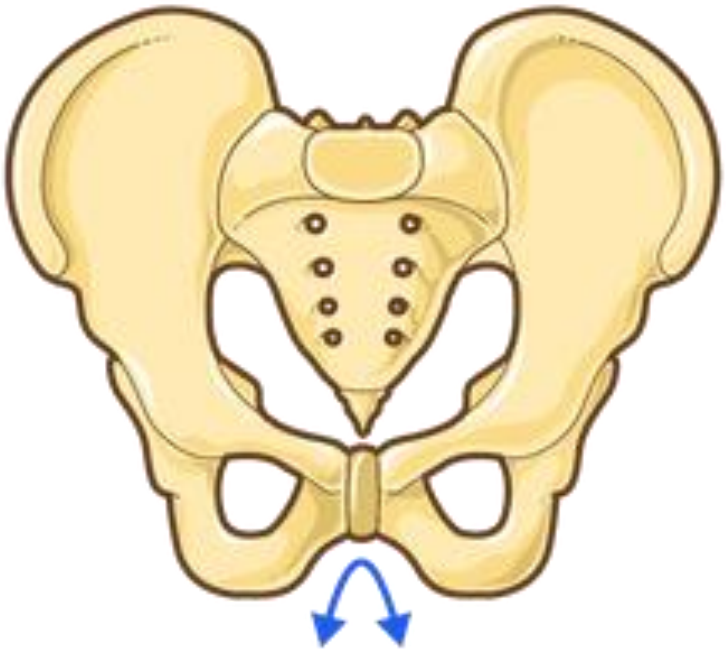
- Iliac crest, ASIS, PSIS
- Ischial spine, ischial tuberosity
- Pubic crest, pubic tubercle
- Acetabulum, obturator foramen
- Sacral promontory, pelvic brim



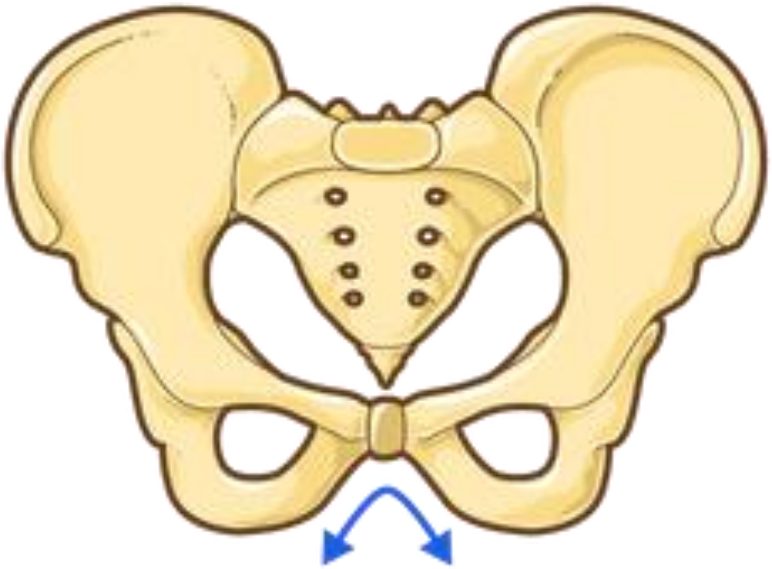


Sex determination of Pelvis

- General build: Male = thick, Female = light
- Pelvic inlet: Male = heart-shaped, Female = oval/round
- Subpubic angle: Male <70°, Female >90°
- Greater sciatic notch: Male = narrow, Female = wide
- Sacrum: Male = long/narrow, Female = short/broad

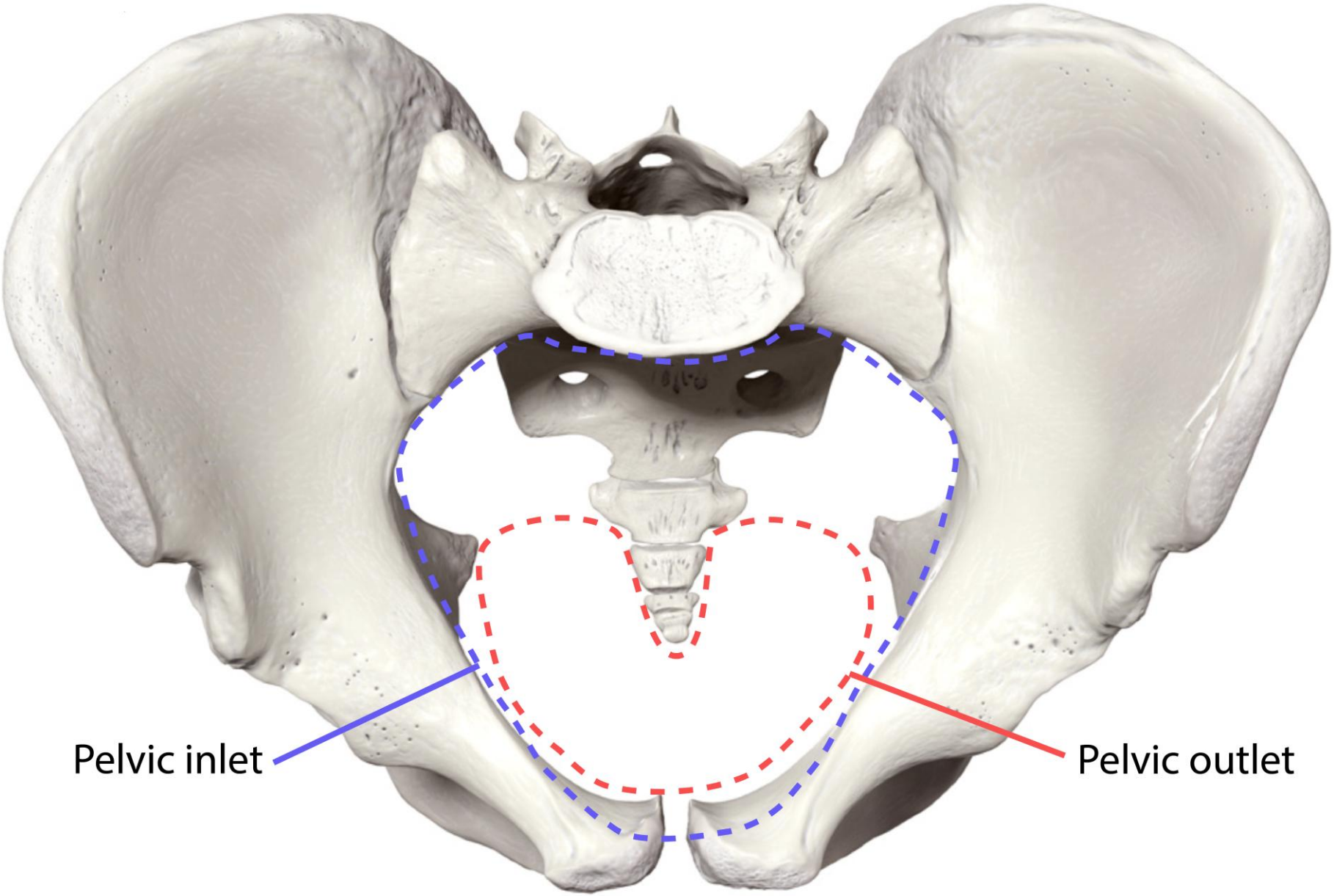


Male



Female

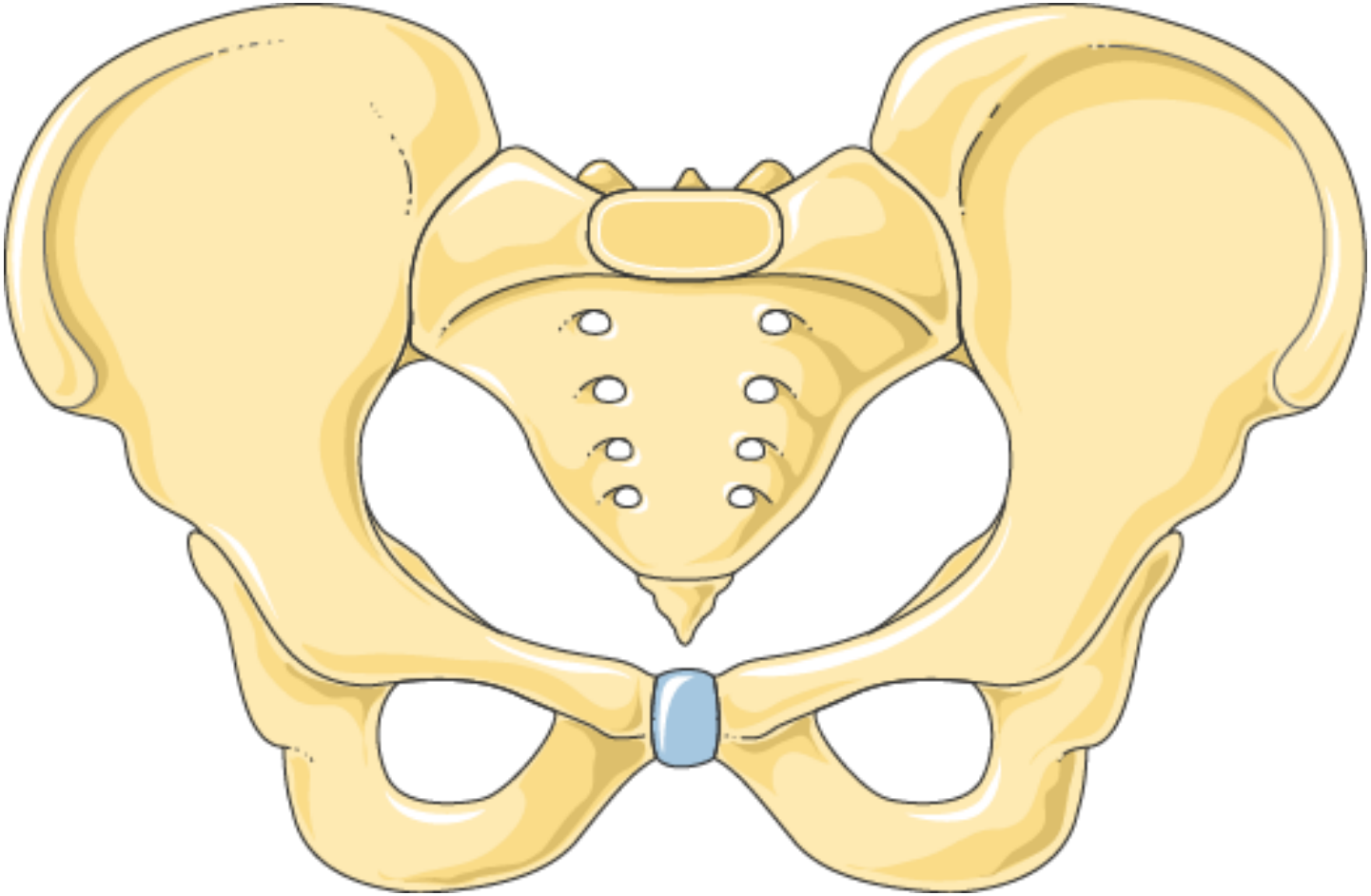
Male Pelvis



Male Pelvis-Key features

- **Pelvic Inlet:** Narrower and heart-shaped, which facilitates weight-bearing and stability.
- **Pelvic Outlet:** Also narrower than in females, to support a more robust body structure.
- **Bony Structure:** Thicker, heavier, and more robust than the female pelvis to provide greater support and stability.
- **Pubic Arch:** More acute angle (less than 90 degrees), also known as the subpubic angle, which is narrow and V-shaped.
- **Sacrum:** Longer, narrower, and more curved, contributing to the overall structural integrity of the pelvis.
- **Iliac Crests:** Higher than in females, making the false pelvis appear taller and narrower.

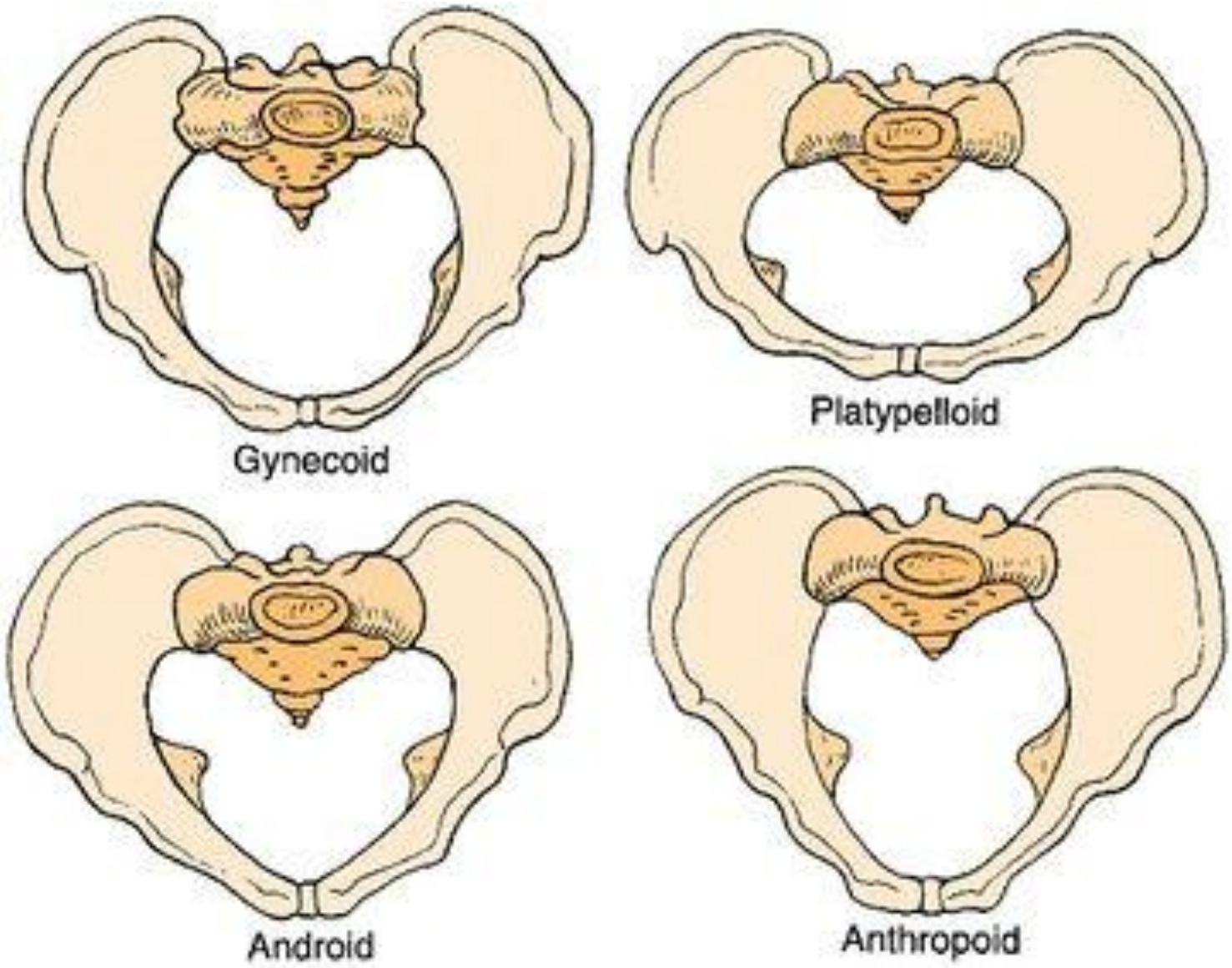
Female Pelvis



Female Pelvis-Key features

- **Pelvic Inlet:** Wider and more oval-shaped in females, providing a larger opening.
- **Subpubic Angle:** The angle formed by the pubic bones is broader in females (around 90-100 degrees), while it is narrower in males.
- **Sciatic Notch:** A wider, U-shaped notch in the female pelvis.
- **Sacrum:** Shorter, broader, and less curved in females, allowing for greater mobility and a wider birth canal.
- **Coccyx:** More mobile in females.
- **Pelvic Cavity:** The true pelvis is the space within the pelvic brim (pelvic inlet) and outlet, which contains the reproductive organs like the uterus, fallopian tubes, and ovaries.

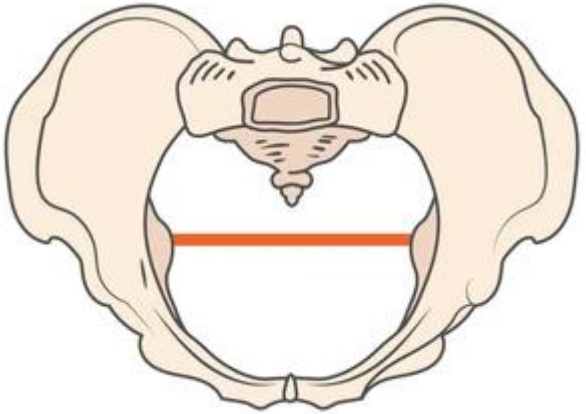
Female Pelvis-Types



Female Pelvis-Gynecoid Type

Key Characteristics

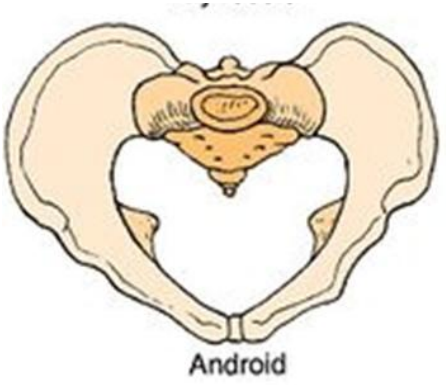
- Pelvic Inlet: opening of the pelvis is oval-shaped, wider than it is long.
- Subpubic Angle:angle formed by the pubic bones is wide and rounded, typically more than 80-90 degrees.
- Sacrum: is shorter and more curved, with a less prominent sacral promontory
- Ischial Spines: e ischium bone) are less prominent, creating a wider "bispinous diameter" which is the distance between the two ischial spines.



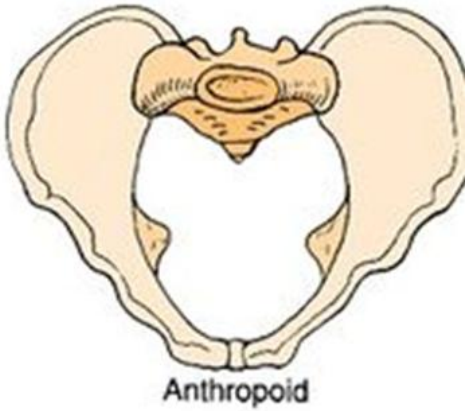
Ideal for childbirth

Female Pelvis-Other Types

Android Pelvis: Tends to be heart-shaped, with a narrower pelvic inlet and less favorable for vaginal birth.



Anthropoid Pelvis: Longer in the anteroposterior (front-to-back) diameter than in the transverse (side-to-side) diameter.



Platypelloid Pelvis: Wide and flat, with a wide transverse diameter but a short anteroposterior diameter.

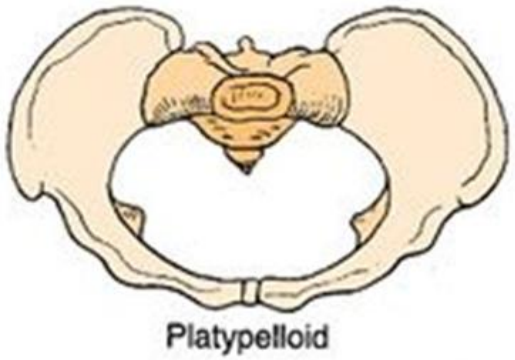
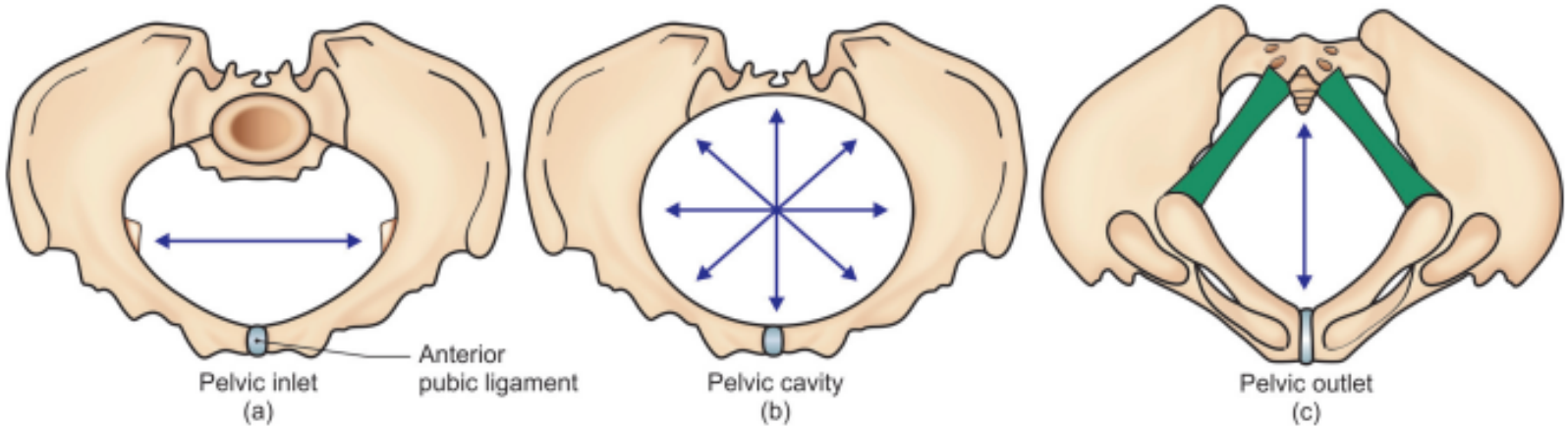


Table 15.1: Dimensions of the pelvis in female (Figs 15.13a to c)

Region	Diameter	Average measurements in cm
I. Pelvic inlet (Fig. 15.13a)	1. *Anteroposterior diameter* (true conjugate), from the midpoint of the sacral promontory to the upper border of the pubic symphysis	11
	2. *Transverse diameter* (maximum)	13
	3. *Oblique diameter*, from one iliopubic eminence to the opposite sacroiliac joint	12
II. Pelvic cavity (Fig. 15.13b)	1. *Anteroposterior diameter*, from midpoint of vertebra S3 to the posterior surface of the pubic symphysis	12
	2. *Transverse diameter*	12
	3. *Oblique diameter*, from the lowest point of one sacroiliac joint to the midpoint of the opposite obturator membrane	12
III. Pelvic outlet (Fig. 15.13c)	1. *Anteroposterior diameter,* from the tip of the coccyx to the inferior margin of the pubic symphysis (maximum)	13
	2. *Transverse diameter* (bituberous diameter), between ischial tuberosities	11
	3. *Oblique diameter*, from the midpoint of the sacrotuberous ligament on one side to the junction of the ischiopubic rami on the other side	12



Figs 15.13a to c: The bony pelvis: (a) Inlet with longest transverse diameter, (b) pelvic cavity with all diameters equal, and (c) outlet with longest anteroposterior diameter

1 The false pelvis is *deep in male* and *shallow in female*.

2 Pelvic inlet is *heart-shaped in male* due to jutting forwards of sacral promontory. In *female, it is transversely oval*.

3 Pelvic cavity is *smaller and deeper in male* (Fig. 15.14a inset). It is a long segment of a short cone. *In female, the pelvic cavity is roomier and shallower* (Fig. 15.14b inset). It is short segment of a long cone (insets).

4 The pelvic outlet is smaller with ischial tuberosities turned inside in male. In female, the pelvic outlet is bigger with everted ischial tuberosities.

5 Sacrum is *longer and narrow in male*, while it is *shorter and wider in female* (Fig. 15.14a insets).

6 *Subpubic angle* is narrower, i.e. *50°–60°, in male*. The angle is wider, i.e. *80°–85°, in female*. This is the most important difference (Figs 15.14a and b).

7 The greater sciatic notch is wider in females (75°) than in males (50°).

8 The acetabulum is large in males, and its diameter is approximately equal to the distance from its anterior margin to the pubic symphysis.

9 The chilotic line extends from the iliopubic eminence to the iliac crest. In females, the pelvic part of the chilotic line is longer than the sacral part.

10 The preauricular sulcus is more marked in females.

Clinical & Forensic Relevance

- Obstetrics: Assessment of pelvic inlet & outlet
- Orthopedics: Pelvic fractures
- Forensic medicine: Sex determination from skeletal remains

Homework

- HOW YOU GOT TO DETERMINE IF THAT IS A PELVIS OF NULLIPAROUS OR PAROUS FEMALE?
- NAME THE STRUCTURES LYING ON THE ALA OF SACRUM
- HOW DO YOU DIFFERENTIATE THE SEX OF SACRUM
- HOW MUCH IS THE SUBPUBIC ANGLE IN MALE AND IN FEMALE PELVIS
- WHICH DIAMETER IS BIGGEST AT THE PUTLET OF THE TRUE PELVIS


Thank you